

Article

Research on Predefined Time Sliding Mode Control Method for High-Speed Maglev Train Based on Finite Time Disturbance Observer

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Abstract: In order to improve the operation control performance of high-speed maglev trains, an improved finite-time rotor magnetic Field-Oriented Control method was proposed in this paper. Aiming at the stator current control problem of long-stator linear synchronous motors under parametric perturbation, this paper investigates the double-feeding mode, combines the predefined-time stability theory and designs an improved sliding mode controller to optimise the dynamic characteristics of the inner-loop system. In the outer-loop cruise control, the predefined-time sliding mode control is combined with a finite-time disturbance observer, which effectively solves the problems of inaccurate modelling and parameter ingestion. It was verified through simulation and analysis that the control strategy has significant advantages in improving the dynamic tracking performance and anti-interference ability, with the stator current stabilisation time within 0.1 s, the absolute value of the fluctuation error within 20 A, the outer-loop response time within 0.5 s, the maximum speed error within 0.0005 m/s and the maximum displacement error within 0.0005 m. The control strategy has the advantages of improving the dynamic tracking performance and anti-interference ability.

Keywords: predefined-time control scheme; finite-time disturbance observer; parametric perturbation; high-speed maglev; sliding mode control



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1. Introduction

As the global population continues to grow, the demand for capacity in the railway transport system is increasing rapidly. High-speed maglev trains, as a cutting-edge mode of transport, have demonstrated broad application prospects due to their advantages of high speed, low operating costs and environmental friendliness. Electromagnetic suspension (EMS)-type maglev trains driven by long-stator linear synchronous motors (LSMs) differ significantly from traditional wheel-rail systems in that the propulsion system is not set up in the train but is arranged along the track, generating thrust through non-contact electric power [1,2]. This technological architecture not only enhances the speed and stability of the train, but also significantly reduces the wear and tear on the track and the train body, thus optimising the long-term maintenance cost and operational efficiency of the system.

When working in double-feeding (DF) mode, the maglev train is a strongly coupled, multiple-input multiple-output system in the dq reference frame, with severe dq-axis current coupling, and each converter current is not only related to its own current and voltage cross-coupling terms but also related to the current and voltage of another converter and the motor excitation current [3–5]. With the increase of speed and the change of motor

parameters during operation, the coupling becomes more and more serious, which is not conducive to the stable operation of the maglev train. The existence of coupling makes the stator current produce spikes [6,7], so it is necessary to investigate the traction control strategy in DF mode.

In order to achieve high-speed operation, maglev trains mostly adopt a DF model with two high-power converters for traction power supply. The LSM behaves as a multi-state, strongly coupled, complex nonlinear system in the dq reference frame, which, due to its unique structure and operating characteristics, puts forward higher requirements on the inner- and outer-loop controllers during high-speed and stable operation. The levitation and propulsion systems of the LSM system share a common air-gap magnetic field, which is a combination of the dynamic magnetic field and the stator armature reaction force, leading to mutual coupling between the two converter currents and voltages and the motor excitation current [8]. As the vehicle speed increases, the motor parameter perturbation causes the current loop coupling to become more and more severe, introducing a large number of current harmonics. The cross-coupling term of the current loop and the model uncertainty of the speed loop seriously affect the dynamic and static control performance of the system. In order to decouple the thrust and suspension forces of the LSM, the rotor Field-Oriented Control method (FOC) with $i_d = 0$ is usually adopted. The traction power supply system of the high-speed maglev train is located on the ground, and the traction method is different from that of the traditional wheel-rail train. The output voltage of the trackside system makes the speed loop control output limited, which seriously affects the performance of the speed loop controller, so it is necessary to investigate the high-precision control method of the maglev train in DF mode.

Current loop control, as the core of high-speed maglev train motor control, needs to have high robustness and fast response speed. Although the traditional PI control has the advantages of easy parameter adjustment and simple structure, it has the disadvantage of contradictory tracking performance and immunity performance, which cannot meet the demand of fast response tracking of the current. References [9–11] used a current control method based on improved Active Disturbance Rejection Control to improve the robustness of the system, but there are problems such as difficult parameter tuning and complex algorithm design. In the train operation composition, the parameter ingestion has a large impact on the dynamic performance of the system. References [12,13] used model-predictive control to improve the transient characteristics of the system, but this method relies on the accuracy of the mathematical modelling of the motor, while increasing the computational burden of the controller. References [14–16] designed an improved sliding mode controller to improve the current tracking accuracy by designing an adaptive mechanism to suppress the fluctuation of the traction force, but the method has poor anti-interference performance, especially when the modelling is inaccurate, and needs to be combined with other algorithms to improve its robustness.

For the velocity loop control, ref. [17] designed an adaptive sliding mode with a fast convergence rate, which ensures a large switching gain and at the same time weakens the time lag phenomenon in the high-frequency dynamics and improves the transient characteristics of the system. Reference [18] proposed a non-singular, fast terminal sliding mode control strategy, which well solves the control problem under the influence of actuator failure and internal and external perturbations, and the strategy combines the sliding mode control with finite-time stability primitives, which makes the error convergence time be visualised, and improves the transient characteristics of the system. However, its error convergence time is related to the initial value of the system. There are many uncertainty parameters, and it is difficult to get the accurate initial value, which makes the finite-time stability theory unable to meet the demands of complex working conditions.

Compared with finite- and fixed-time stability, the error convergence time of predefined time control is independent of the initial state of the system [19], and the accuracy of the convergence time range is improved by using adjustable parameters of the controller as an upper bound of time. Based on the above discussion, this paper proposes a predefined time sliding mode control method based on a finite-time disturbance observer for the control requirements of high-speed maglev trains, combined with the predefined time stability theorem. Specifically, the contributions of this paper include the following:

- (1) The theory of predefined time is introduced into the design of improved sliding mode controllers for high-speed maglev trains. A new predefined-time sliding mode control structure is designed, which is applied to the inner-loop stator current and the outer-loop speed error control, to achieve the adjustability of the upper bound of the convergence time and, at the same time, make the upper bound free from the influence of any other parameter and the initial conditions, so as to improve the system's immunity to disturbances.
- (2) The inner and outer loop uncertainty terms such as complex unknown disturbances and additional resistance of the system are estimated as aggregate disturbances, which are estimated and compensated by using a finite-time disturbance observer, which improves the observation accuracy and outer-loop robustness while satisfying the stability to eliminate the impact of this part of the disturbance on the operation of the high-speed maglev train.
- (3) The designed control strategy simultaneously considers the effects of the LSLSM parameter perturbation, uncertain resistance and unknown perturbations. In addition, compared to other uncertain uptake estimation methods that only assume that the parameters are unknown [20,21], this paper makes the more general assumption that all unknown parameters are time-varying. This extends the range of practical applications while achieving global finite-time stability.

The rest of the paper is organised as follows. In Section 2, the paper develops a mathematical model of a long-stator linear synchronous motor in DF mode and analyses the dynamics of the system in detail. In Section 3, some useful theorems are introduced to explain the design process of the proposed control strategy. In Section 4, the effectiveness of the proposed method is verified through simulation experiments, demonstrating its feasibility and advantages in practical applications. Finally, the main research results are summarised in Section 5.

2. Dynamic Model Analysis

2.1. LSM Voltage Equation

In establishing the mathematical model of a high-speed maglev train, it is first necessary to analyse in depth the characteristics of the individual converter and its interrelationship with the stator windings. The mathematical model of a long-stator linear synchronous motor (LSM), although similar to a conventional rotating motor in its principle of operation, includes not only the dq-axis voltages of the converter u_d and u_q , the dq-axis stator currents i_d and i_q , mover electrical angular speed ω , the inductances of the feeder cable L_k , the resistances of the feed cable R_k , total stator resistance R_s , the dq-axis stator inductance L_d and L_q and pole distance τ , but also the key parameters, such as the excitation current i_m and the mutual inductance M_{sm} . The combined effect of these parameters determines the electromagnetic characteristics and dynamic behaviour of the system. The model of the LSM in the dq-axis reference system can be expressed as:

$$\begin{bmatrix} u_d \\ u_q \end{bmatrix} = \begin{bmatrix} R_s + R_k + pL_d + pL_k & -\omega L_q - \omega L_k \\ \omega L_d + \omega L_k & R_s + R_k + pL_q + pL_k \end{bmatrix} \times \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ \omega M_{sm} i_m \end{bmatrix} \quad (1)$$

When the train operates at low speeds, the system is usually in the single-feeding (SF) mode, the stator section is powered by only one converter. Due to the limited capacity of a single converter, it is difficult to achieve high-speed operation of maglev trains in the SF mode. In order to meet the demand for high-speed operation, the DF mode is often used, where two converters are used to drive the vehicle at the same time. The DF mode not only provides sufficient thrust to ensure the performance of the vehicle during the high-climbing and acceleration phases but also reduces the complexity of the design and fabrication as the design capacity of each converter is relatively small. More critically, the DF mode enhances the safety redundancy of the system, so that even if one converter fails, the other converter is still able to maintain power to the stator section. In addition, since the stator current is distributed between the two converters, this significantly reduces the power losses in the feeder cable system and improves the overall efficiency of the system. The schematic diagram of the long-stator LSM in DF mode is shown in Figure 1.

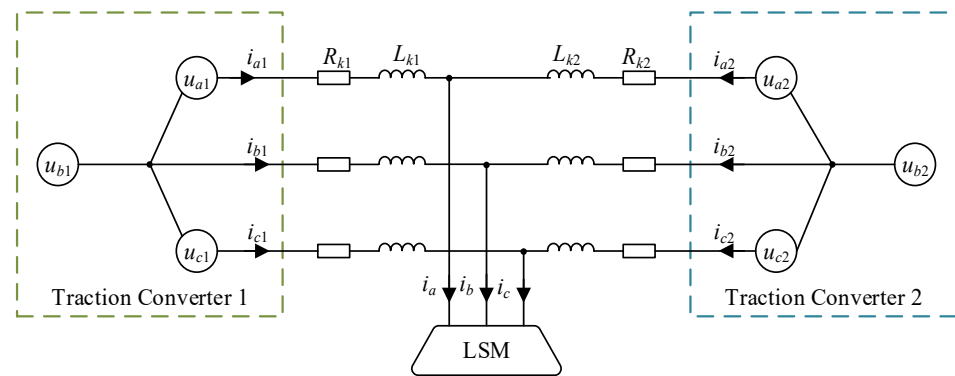


Figure 1. Equivalent circuit diagram of the LSM in DF mode.

In the DF mode, the mathematical modelling of the LSM system, which can be represented as a complex strongly coupled, nonlinear and multiple variable system in the DF mode, becomes more complicated. Especially in DF mode, the coupling problem of the dq-axis currents becomes prominent. Each converter not only has its own current and voltage coupling but also has external coupling with the current and voltage of the other converter and the motor excitation current.

In order to accurately describe the system behaviour, the two converters can be considered as a whole. The state equations of the system can be derived by taking the sum of the output voltages and the difference of the voltages of the converters as the input variables, and the currents and circulating currents jointly output by both of them as the state variables [4]. The model is transformed to simplify the coupling terms, which not only considers the internal coupling terms related to the motor parameters and excitation currents, as well as those related to the inductance and resistance of the feeder cables, but also reflects the complex cross-coupling relationship between the two converters. In the DF mode, the total stator current of the LSM is larger while the circulating current is smaller [5]. We took the rotor coordinate system (d-q) as the reference coordinates, attributed the effects

brought by the system circulating current and parameter ingress to perturbations and derived the equations of the state of the inner loop, given by Equations (2) and (3):

$$\begin{bmatrix} u_{dsum} \\ u_{qsum} \\ u_{dsub} \\ u_{qsub} \end{bmatrix} = \begin{bmatrix} L_{dz}p + \frac{R_{sum}}{2} + 2R_s & -\omega L_{qz} & L_{sub}p + R_{sub} & -\omega L_{sub} \\ \omega L_{dz} & L_{qz}p + \frac{R_{sum}}{2} + 2R_s & \omega L_{sub} & L_{sub}p + R_{sub} \\ \frac{L_{sub}p}{2} + \frac{R_{sub}}{2} & -\frac{\omega L_{sub}}{2} & L_{sum}p + R_{sum} & -\omega L_{sum} \\ \frac{\omega L_{sub}}{2} & \frac{L_{sub}p}{2} + \frac{R_{sub}}{2} & \omega L_{sum} & L_{sum}p + R_{sum} \end{bmatrix} \begin{bmatrix} i_{dsum} \\ i_{qsum} \\ i_{dsub} \\ i_{qsub} \end{bmatrix} + \begin{bmatrix} 0 \\ 2\omega M_{sm} i_m \\ 0 \\ 0 \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} u_{dsum} \\ u_{qsum} \\ i_{dsum} \\ i_{qsum} \end{bmatrix} = \begin{bmatrix} u_{d1} + u_{d2} \\ u_{q1} + u_{q2} \\ i_{d1} + i_{d2} \\ i_{q1} + i_{q2} \end{bmatrix}, \quad \begin{bmatrix} u_{dsub} \\ u_{qsub} \\ i_{dsub} \\ i_{qsub} \end{bmatrix} = \begin{bmatrix} u_{d1} - u_{d2} \\ u_{q1} - u_{q2} \\ (i_{d1} - i_{d2})/2 \\ (i_{q1} - i_{q2})/2 \end{bmatrix}, \quad \begin{bmatrix} L_{dz} \\ L_{qz} \\ R_{sum} \\ R_{sub} \\ L_{sum} \\ L_{sub} \end{bmatrix} = \begin{bmatrix} \frac{L_1 + L_2}{2} + 2L_d \\ \frac{L_1 + L_2}{2} + 2L_q \\ R_1 + R_2 \\ R_1 - R_2 \\ L_1 + L_2 \\ L_1 - L_2 \end{bmatrix} \quad (3)$$

where L_{k1} and L_{k2} represent the inductance of the feeder cables on two sides; R_{k1} and R_{k2} denote its resistance; u_{d1} and u_{q1} correspond to the dq-axis output voltages of converter 1, while i_{d1} and i_{q1} are its currents; u_{d2} and u_{q2} denote the dq-axis output voltages of converter 2, and i_{d2} and i_{q2} represent its currents; u_{dsum} , u_{qsum} , i_{dsum} and i_{qsum} refer to the sum of the output voltages and currents of the two converters in the d-q axis; u_{dsub} , u_{qsub} , i_{dsub} and i_{qsub} represent the difference in output voltages and the circulating current component between the two converters in the d-q axis. As shown in Equation (2), the sum of the output voltages is influenced by its cross-coupling currents and voltages, imposing stricter requirements on achieving stable control of the stator current.

2.2. Mechanical Model

In this section, the train is equivalently modeled as an LSM motor for force analysis. During the modeling of longitudinal dynamics, the train can be approximated as a mass point influenced by traction/braking forces, basic resistance, additional resistance and unknown disturbances [22]. Based on Newton's laws of motion, the mechanical model of a high-speed maglev train as a single mass point is expressed as shown in Equation (4):

$$\begin{cases} \dot{x} = v \\ \dot{v} = (F_x - F_z)/m + f \\ F_x = \frac{3\pi}{2\tau} M_{sm} i_m i_q^* + \frac{3\pi}{2\tau} (L_d - L_q) i_d^* i_q^* \\ F_z = N(a\sqrt{v} + bv^{0.7} + cv^2) + F_b \\ F_b = \begin{cases} 0 & v \in [0-20]\text{km/h} \\ N \cdot 7300 & v \in [20-70]\text{km/h} \\ N \cdot (146/v - 0.2) \cdot 1000 & v \in [70-600]\text{km/h} \end{cases} \end{cases} \quad (4)$$

where F_x represents the electromagnetic thrust of the train, F_z denotes the total resistance comprising aerodynamic resistance, magnetic resistance and other components, and F_b refers to the engine resistance, x and v correspond to the train's position and velocity, respectively, f represents the unknown disturbance acting on the train, and a , b and c are the train's resistance parameters.

Neglecting torque ripple caused by Field-Oriented Control and setting the reference current i_d^* to zero [8], the differential torque can be derived from Equation (4):

$$\ddot{x}_i = \frac{3\pi M_{sm} i_m}{2\tau m} i_q^* - \frac{F_z}{m} + f \quad (5)$$

To simplify the calculations, the parameters were defined as follows:

$$M = \frac{2\tau m}{3\pi M_{sm} i_m}, b = \frac{2\tau}{3\pi M_{sm} i_m} \quad (6)$$

Let $i_d^* = u$ be the output of the velocity loop controller. Equation (4) can then be rewritten as:

$$\begin{cases} \dot{x} = v \\ M\dot{v} = u - bF_z + Mf \end{cases} \quad (7)$$

In the practical operation of maglev trains, various unknown factors, such as the complexity of the train body structure and fluctuations in power supply, can affect parameter values, making it challenging to determine these parameters accurately [23]. To validate the generality of the control framework, this study models the resistance coefficient as an unknown time-varying function, consisting of an unknown time-varying component and an unknown constant component:

$$\begin{cases} a = a^s + \Delta a \\ b = b^s + \Delta b \\ c = c^s + \Delta c \end{cases} \quad (8)$$

where a^s , b^s and c^s are unknown constant parameters, and Δa , Δb and Δc are unknown time-varying parameters.

Assumption 1 ([24]). The following sequence describes the unknown time-varying component of the train resistance parameters, whose variation is bounded by an unknown constant limit:

$$|\Delta a| \leq a^{\sup}, |\Delta b| \leq b^{\sup}, |\Delta c| \leq c^{\sup} \quad (9)$$

where $a^{\sup}$, $b^{\sup}$ and $c^{\sup}$ are unknown constant parameters.

Assumption 2. For the model described by Equation (5), the maglev train's actuator is equipped with a feasible controller capable of achieving the desired system performance, and the disturbance term f is a bounded variable.

3. Controller Design

In this section, a predefined-time, improved sliding mode observer-compensation control scheme is proposed and applied to the operational control design of high-speed maglev trains with a dual-loop structure. All * are reference variables. The control architecture is illustrated in Figure 2. An improved nonsingular sliding mode structure is designed based on predefined-time stability theory and is implemented in both the inner loop for stator current control and the outer loop for velocity control. This ensures that the system states converge to the desired stable state within the predefined time.

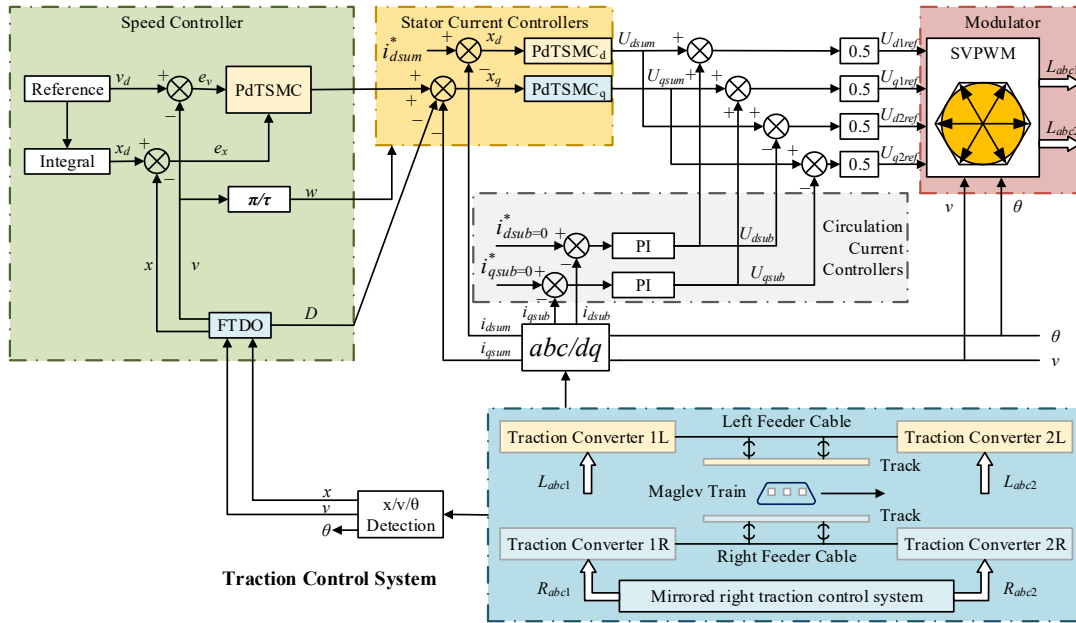


Figure 2. Block diagram of the dual-loop control system.

3.1. Relevant Mathematical Lemmas

Lemma 1 ([24]). *For the following system:*

$$\begin{aligned} \dot{\sigma}_0 &= -\lambda_0 L^{1/(n+1)} |\sigma_0|^{n(n+1)} \text{sgn}(\sigma_0) + \sigma_1 \\ \dot{\sigma}_1 &= -\lambda_1 L^{1/n} |\sigma_1 - \dot{\sigma}_0|^{(n-1)/n} \text{sgn}(\sigma_1 - \dot{\sigma}_0) + \sigma_2 \\ &\vdots \\ \dot{\sigma}_{n-1} &= -\lambda_{n-1} L^{1/2} |\sigma_{n-1} - \dot{\sigma}_{n-2}|^{1/2} \text{sgn}(\sigma_{n-1} - \dot{\sigma}_{n-2}) + \sigma_n \\ \dot{\sigma}_n &\in -\lambda_n L \text{sgn}(\sigma_n - \dot{\sigma}_{n-1}) + [-L, L] \end{aligned} \quad (10)$$

where $L > 0$ and $\lambda_i (i = 0, 1, \dots, n)$ are both positive constants. The system can achieve finite-time stability by selecting appropriate parameters.

Lemma 2 ([19]). *For a constant $T > 0$, if there exists an unbounded Lyapunov function $V(x)$ for system $\dot{x}(t) = \phi(t, x)$, the following equation is satisfied:*

$$\dot{V} \leq -\frac{\pi}{\alpha T_c} \left(V^{1-\frac{\alpha}{2}} + V^{1+\frac{\alpha}{2}} \right) \quad (11)$$

Then, $\dot{x}(t) = \phi(t, x)$ is defined as predefined-time stable, and $\alpha \in (0, 1)$.

3.2. Controller Design and Stability Analysis

This section addresses the dual-loop FOC vector control problem for high-speed maglev trains by analysing the state equations of the inner and outer loops. The design of the PdTSMC and the finite-time disturbance observer is conducted, followed by a stability analysis of the proposed novel control structure.

3.2.1. Predefined-Time Sliding Mode Controller

For the dual-loop control system, a predefined-time sliding mode controller (PdTSMC) is designed. The tracking error of the d-q axis stator current is first treated as the inner-loop

state variable, while the train speed error and displacement error are considered as the outer-loop state variables, as shown in Equation (12):

$$\begin{bmatrix} x_d \\ x_q \\ x_v \\ x_s \end{bmatrix} = \begin{bmatrix} i_{dsum}^* - i_{dsum} \\ i_{qsum}^* - i_{qsum} \\ v_{ref} - v \\ s_{ref} - s \end{bmatrix} \quad (12)$$

In the DF mode, the total stator current of the LSM is large, while the circulating current is small [8]. Using the dq-axis as the reference frame, the effects of system circulating current and parameter perturbations are treated as disturbances. The inner-loop state is derived as Equations (13) and (14):

$$\begin{bmatrix} u_{dsum} \\ u_{qsum} \end{bmatrix} = \begin{bmatrix} pL_{dz} + R_s & -\omega L_{qz} \\ \omega L_{dz} & pL_{qz} + R_s \end{bmatrix} \begin{bmatrix} i_{dsum} \\ i_{qsum} \end{bmatrix} + \begin{bmatrix} f_d \\ f_q + 2\omega M_{sm} i_m \end{bmatrix} \quad (13)$$

$$\begin{cases} f_d = \Delta R_s i_{dsum} + \Delta L_{dz} \frac{d}{dt} i_{dsum} - \omega \Delta L_{qz} i_{qsum} + \delta_d \\ f_q = \Delta R_s i_{qsum} + \Delta L_{dz} \frac{d}{dt} i_{qsum} + \omega (\Delta L_{dz} i_{dsum} + \Delta M_{sm} i_m) + \delta_q \end{cases} \quad (14)$$

where δ_d and δ_q represent the circulating current and other uncertainty disturbances, Δ is the parameter uncertainty component, and f_d and f_q are the inner-loop disturbance components.

For the inner-loop error system, consider the state variables x_d and x_q :

$$\begin{cases} \dot{x}_{d1} = x_{d2} = i_{dsum}^* - i_{dsum} \\ \dot{x}_{d2} = \frac{1}{L_{dz}} (R_s i_{dsum} - \omega L_{qz} i_{qsum} - u_{dsum}) \end{cases} \quad (15)$$

$$\begin{cases} \dot{x}_{q1} = x_{q2} = i_{qsum}^* - i_{qsum} \\ \dot{x}_{q2} = \frac{1}{L_{qz}} (R_s i_{qsum} + \omega L_{qz} i_{dsum} + 2\omega M_{sm} i_m - u_{qsum}) \end{cases} \quad (16)$$

To ensure the error system is stable within predefined time, a novel sliding mode surface is designed as follows:

$$\begin{cases} s_d = x_{d2} + \int_0^t C_{1d} \text{sign}(x_{d2}) |x_{d2}|^{1-\alpha_d} + C_{2d} \text{sign}(x_{d2}) |x_{d2}|^{1+\alpha_d} + C_{3d} \text{sign}(x_{d2}) d\tau \\ s_q = x_{q2} + \int_0^t C_{1q} \text{sign}(x_{q2}) |x_{q2}|^{1-\alpha_q} + C_{2q} \text{sign}(x_{q2}) |x_{q2}|^{1+\alpha_q} + C_{3q} \text{sign}(x_{q2}) d\tau \end{cases} \quad (17)$$

Taking the derivative of the sliding mode surface in Equation (17) yields the following:

$$\begin{cases} \dot{s}_d = \dot{x}_{d2} + C_{2d} \text{sign}(x_{d2}) |x_{d2}|^{1-\alpha_d} + C_{2d} \text{sign}(x_{d2}) |x_{d2}|^{1+\alpha_d} + C_{3d} \text{sign}(x_{d2}) \\ \dot{s}_q = \dot{x}_{q2} + C_{2q} \text{sign}(x_{q2}) |x_{q2}|^{1-\alpha_q} + C_{2q} \text{sign}(x_{q2}) |x_{q2}|^{1+\alpha_q} + C_{3q} \text{sign}(x_{q2}) \end{cases} \quad (18)$$

A control law u is designed to ensure that the stator current error converges to the sliding mode surface within the predefined time:

$$\begin{cases} u_{dsum} = \omega L_{qz} i_{qsum} - R_s i_{dsum} + L_{dz} \left[C_{2d} \text{sign}(x_{d2}) |x_{d2}|^{1-\alpha_d} + C_{2d} \text{sign}(x_{d2}) |x_{d2}|^{1+\alpha_d} + C_{3d} \text{sign}(x_{d2}) \right. \\ \quad \left. + C_{4d} \text{sign}(s_d) |s_d|^{1-\alpha_d} + C_{5d} \text{sign}(s_d) |s_d|^{1+\alpha_d} + C_{6d} \text{sign}(s_d) \right] \\ u_{qsum} = \omega L_{dz} i_{dsum} + R_s i_{qsum} + L_{qz} \left[C_{2q} \text{sign}(x_{q2}) |x_{q2}|^{1-\alpha_q} + C_{2q} \text{sign}(x_{q2}) |x_{q2}|^{1+\alpha_q} + C_{3q} \text{sign}(x_{q2}) \right. \\ \quad \left. + C_{4q} \text{sign}(s_q) |s_q|^{1-\alpha_q} + C_{5q} \text{sign}(s_q) |s_q|^{1+\alpha_q} + C_{6q} \text{sign}(s_q) \right] \end{cases} \quad (19)$$

where $\alpha \in (0, 1)$, $c_{1d} = \frac{\pi}{\alpha T_1} \left(\frac{1}{2}\right)^{1-\frac{\alpha_d}{2}}$, $c_{2d} = \frac{\pi}{\alpha T_1} \left(\frac{1}{2}\right)^{1+\frac{\alpha_d}{2}}$, $c_{4d} = \frac{\pi}{\alpha T_2} \left(\frac{1}{2}\right)^{1-\frac{\alpha_d}{2}}$, $c_{5d} = \frac{\pi}{\alpha T_2} \left(\frac{1}{2}\right)^{1+\frac{\alpha_d}{2}}$, $c_{1q} = \frac{\pi}{\alpha T_1} \left(\frac{1}{2}\right)^{1-\frac{\alpha_q}{2}}$, $c_{2q} = \frac{\pi}{\alpha T_1} \left(\frac{1}{2}\right)^{1+\frac{\alpha_q}{2}}$, $c_{4q} = \frac{\pi}{\alpha T_2} \left(\frac{1}{2}\right)^{1-\frac{\alpha_q}{2}}$, $c_{5q} = \frac{\pi}{\alpha T_2} \left(\frac{1}{2}\right)^{1+\frac{\alpha_q}{2}}$, and $c_{3d}, c_{6d}, c_{3q}, c_{6q} > 0$.

Similarly, for the outer-loop state variables x_v and x_s , the predefined-time sliding mode surface is designed as follows:

$$s_d = x_v + \int_0^t C_1 \text{sign}(x_v) |x_v|^{1-\lambda} + C_2 \text{sign}(x_v) |x_v|^{1+\lambda} + C_3 \text{sign}(x_v) d\tau \quad (20)$$

The control law u is designed as:

$$u = M \left[C_1 \text{sign}(x_v) |x_{d2}|^{1-\lambda} + C_2 \text{sign}(x_v) |x_v|^{1+\lambda} + C_3 \text{sign}(x_v) + C_4 \text{sign}(s_v) |s_v|^{1-\lambda} + C_5 \text{sign}(s_v) |s_v|^{1+\lambda} + C_6 \text{sign}(s_v) \right] + \frac{F_z}{m} \quad (21)$$

where $\beta \in (0, 1)$, $c_1 = \frac{\pi}{\alpha T_3} \left(\frac{1}{2}\right)^{1-\frac{\lambda}{2}}$, $c_2 = \frac{\pi}{\alpha T_3} \left(\frac{1}{2}\right)^{1+\frac{\lambda}{2}}$, $c_3 > 0$, $c_4 = \frac{\pi}{\alpha T_4} \left(\frac{1}{2}\right)^{1-\frac{\lambda}{2}}$, $c_5 = \frac{\pi}{\alpha T_4} \left(\frac{1}{2}\right)^{1+\frac{\lambda}{2}}$, $c_6 > 0$.

To address the impacts of modeling uncertainties and unknown disturbances, this study considers the aggregate disturbance D , which combines the unknown time-varying resistance components, the train's unknown disturbance f and the parameter perturbation-induced interference from the inner-loop circuit. Based on the outer-loop state equations, a finite-time disturbance observer is designed to provide feedforward compensation for these disturbances:

$$\begin{aligned} \dot{\hat{v}} &= \text{sat}(u) - f + \dot{z}_0 \\ \dot{z}_0 &= -\beta_0 L^{1/3} |\hat{v} - v|^{2/3} \text{sgn}(\hat{v} - v) + z_1 \\ \dot{z}_1 &= -\beta_1 L^{1/2} |z_1 - \dot{z}_0|^{1/2} \text{sgn}(z_1 - \dot{z}_0) + z_2 \\ \dot{z}_2 &= -\beta_2 L \text{sgn}(z_2 - \dot{z}_1) \end{aligned} \quad (22)$$

where $\hat{v}$, $\hat{D}$ and $\hat{\dot{D}}$ represent the estimates of z_0 , z_1 and z_2 , respectively. The observer is designed in compliance with Lemma 1, ensuring that the observation error converges within a finite time. In order to improve the dynamic performance of the system, here we use only the estimation of the aggregate interference.

3.2.2. Stability Analysis

To verify the stability of the designed controller, the reaching phase ($s \neq 0$) is first considered. A Lyapunov function for the d axis stator current is constructed as follows:

$$V_{2d} = \frac{1}{2} s_d^2 \geq 0 \quad (23)$$

The derivative can be computed from Equation (23):

$$\begin{aligned} \dot{V}_{2d} &= s_d \dot{s}_d \\ &= s \left[\dot{x}_{d2} + C_{1d} \text{sign}(x_{d2}) |x_{d2}|^{1-\alpha_d} + C_{2d} \text{sign}(x_{d2}) |x_{d2}|^{1+\alpha_d} + C_{3d} \text{sign}(x_{d2}) \right] \\ &= -s \left[C_{4d} \text{sign}(s) |s|^{1-\alpha_d} + C_{5d} \text{sign}(s) |s|^{1+\alpha_d} + C_{6d} \text{sign}(s) \right] \\ &= -C_{4d} |s|^{2-\alpha_d} - C_{5d} |s|^{2+\alpha_d} - C_{6d} |s| \\ &\leq -C_{4d} |s|^{2-\alpha_d} - C_{5d} |s|^{2+\alpha_d} \end{aligned} \quad (24)$$

Thus,

$$\dot{V}_{2d} \leq -\frac{\pi}{\alpha T_2} \left(V_{2d}^{1-\frac{\alpha_d}{2}} + V_{2d}^{1+\frac{\alpha_d}{2}} \right) \quad (25)$$

The stator current error will converge along the sliding surface within the predefined time T_2 .

For the sliding phase ($s \equiv 0$), the state variables are confined to the sliding surface $s_d(t) = \{x_i | s(x_i) = 0\}$ and satisfy $\dot{s}_d(x) = 0$. The Lyapunov function is designed as follows:

$$V_{1d} = \frac{1}{2} x_{d2}^2 \geq 0 \quad (26)$$

The derivative is calculated as follows:

$$\begin{aligned} \dot{V}_{1d} &= x_{2d} \dot{x}_{2d} \\ &= -x_{2d} \left[C_{1d} \text{sign}(x_{d2}) |x_{d2}|^{1-\alpha_d} + C_{2d} \text{sign}(x_{d2}) |x_{d2}|^{1+\alpha_d} + C_{3d} \text{sign}(x_{d2}) \right] \\ &= -C_{1d} |x_{d2}|^{2-\alpha_d} - C_{2d} |x_{d2}|^{2+\alpha_d} - C_{3d} |x_{d2}| \\ &\leq -C_{1d} |x_{d2}|^{2-\alpha_d} - C_{2d} |x_{d2}|^{2+\alpha_d} \end{aligned} \quad (27)$$

Therefore,

$$\dot{V}_{1d} \leq -\frac{\pi}{\alpha T_1} \left(V_{1d}^{1-\frac{\alpha_d}{2}} + V_{1d}^{1+\frac{\alpha_d}{2}} \right) \quad (28)$$

Thus, the d-axis stator current error will converge to zero along the sliding surface within the predefined time $T = T_1 + T_2$.

Similarly, the q-axis current and the outer-loop state errors will also converge within the predefined time. The predefined convergence time for the q-axis current error is the same as that for the d-axis current, while the convergence time for the outer-loop stability is $T_o = T_3 + T_4$.

4. Model Validation

To verify the effectiveness and reliability of the proposed improved FOC strategy, this section presents a simulation study using an EMS-type high-speed maglev train as an example. The specific parameters of the train are provided in Table 1. Since the train does not have access to high-speed test tracks, a target trajectory curve is designed in this section. Figure 3 shows the target velocity curve for the leading maglev train. The entire operation time lasted for 380 s. From 0 to 200 s, the reference target curve accelerated with an acceleration of 0.5 m/s^2 , followed by 10 s of cruising at a constant speed, and finally decelerated with a deceleration of 0.6 m/s^2 until it came to a stop.

Table 1. LSM parameters.

Motor Parameters	Value
Stator resistance $R_s(m\Omega)$	436.176
Mutual inductance $M_{sm}(mH)$	165
Stator d-axis inductance $L_d(mH)$	3.9153
Stator q-axis inductance $L_q(mH)$	3.4653
Motor pole pitch $\tau(m)$	0.258
Gross vehicle of train $M(t)$	306.9
Feeder cable resistance $R_{k1}(m\Omega/km)$	58.33
Feeder cable inductance $L_{k1}(mH/km)$	0.0713
Excitation current $I_m(A)$	27
Number of train sections N	5

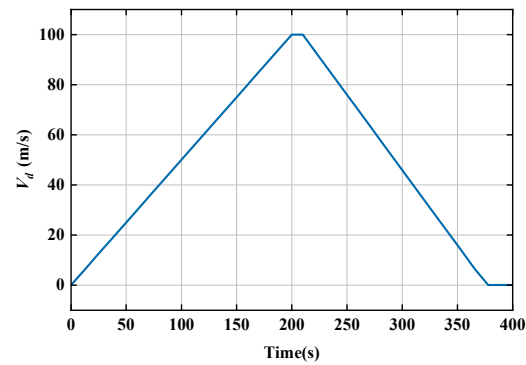


Figure 3. Velocity input curve.

The current loop simulation analysis was performed first. In the experiment, the stator current controller under the DF mode was simulated and validated using three control strategies: voltage feedforward decoupling method (PI-VFDC), linear sliding mode control SMC and predefined-time sliding mode control (PdTSMC), denoted as strategy I, strategy II and strategy III, respectively. The controller parameters for the three strategies are shown in Table 2, and the table of the maximum inner-loop controller performance is shown in Table 3.

Table 2. The current loop controller parameters.

Controller	Parameters	Value
PI-VFDC	K_p	7
	K_i	10
SMC	α_{cd}	1
	α_{cq}	1
	T_{cd}	0.1
	T_{cq}	0.1
	ε_c	10
PdTSMC	α_d	0.8
	α_q	0.001
	T_1	0.1
	T_2	0.1
	$c_{3d/q}$	10
	$c_{6d/q}$	10

Table 3. Comparison of the inner-loop controller performance.

Controller	Maximum Current Error	Response Time
PI-VFDC	174 A	0.8 s
SMC	37 A	0.6 s
PdTSMC	20 A	0.1 s

Considering that maglev trains are highly susceptible to disturbances from the track-side environment during high-speed operation, which can lead to the LSM parameter perturbations, the simulation included the effects of parameter perturbations at 61 s while the current followed the step command shown in Figure 4. The feeder cable resistance and inductance from Table 1 were combined with the inductance of the electrical system, and both were stepped to 2 times and 0.5 times their initial values, respectively, for a duration

of 3.5 s before returning to normal values. The total resistance and total inductance were expressed as:

$$\begin{cases} R_{zs} = R_{k1} + R_{k2} + R_s \\ L_{zs} = L_{k1} + L_{k2} + L_d + L_q \end{cases} \quad (29)$$

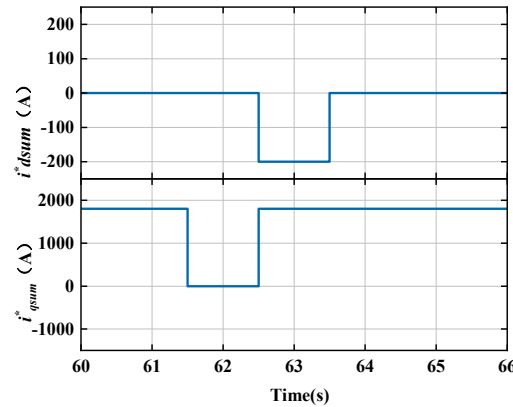


Figure 4. Stator current input curve.

Figure 5 shows the dq-axis stator current curves under the same parameter perturbations for the three strategies. From the step change in current, it can be observed that strategies I and II did not track the abrupt command changes effectively. The traditional PI-VFDC strategy I was highly dependent on the accuracy of motor parameters, and any fluctuation in one axis current caused a change in the current of the other axis, failing to effectively address the current coupling issue. When parameter perturbations occurred, the fluctuations were especially significant. The fluctuation range of the currents i_{dsum} and i_{qsum} was substantial, with the maximum current error reaching 174 A. There was a large delay effect, and the adjustment time was relatively long. The system shows poor anti-interference capability and transient characteristics.

The system employs a feedforward decoupling compensation strategy; however, due to large parameter estimation errors, the decoupling compensation fails to fully cancel out the actual coupling effect. Especially in the feedforward-based control algorithm, the rotational transformation introduces cross-coupling terms, which are not completely decoupled. As a result, the fluctuations in the d-q axis currents cannot be fully eliminated. When the d-q axis currents experience significant fluctuations, the coupling effect becomes more pronounced as the vehicle speed increases, further exacerbating the current fluctuations. In high-speed conditions, this coupling effect causes the amplitude of current fluctuations to exceed 200 A, leading to the oscillatory phenomenon shown in Figure 5.

To demonstrate the superiority of predefined-time sliding mode control, strategy II set $\alpha = 1$ such that the sliding mode surface, as shown in Equation (17), degenerated into a nonfinite-time sliding mode surface. As seen in Figure 5, compared to strategy I, most of the stator current fluctuations caused by parameter perturbations were suppressed. The fluctuation range of i_{dsum} and i_{qsum} was limited to within 30 A, improving the system's disturbance rejection capability. The maximum current error was 37 A, and the dq-axis current response time was confined within 0.6 s. The system was still affected by parameter perturbations, a certain delay effect remained, and a noticeable chattering phenomenon was present. The performance of the inner loop requires further improvement.

Strategy III further constrained the current fluctuation range. As shown in Figure 5, the coupling effect of the dq-axis stator current was almost eliminated. Compared to the other strategies, strategy III effectively mitigated the impact of parameter perturbations.

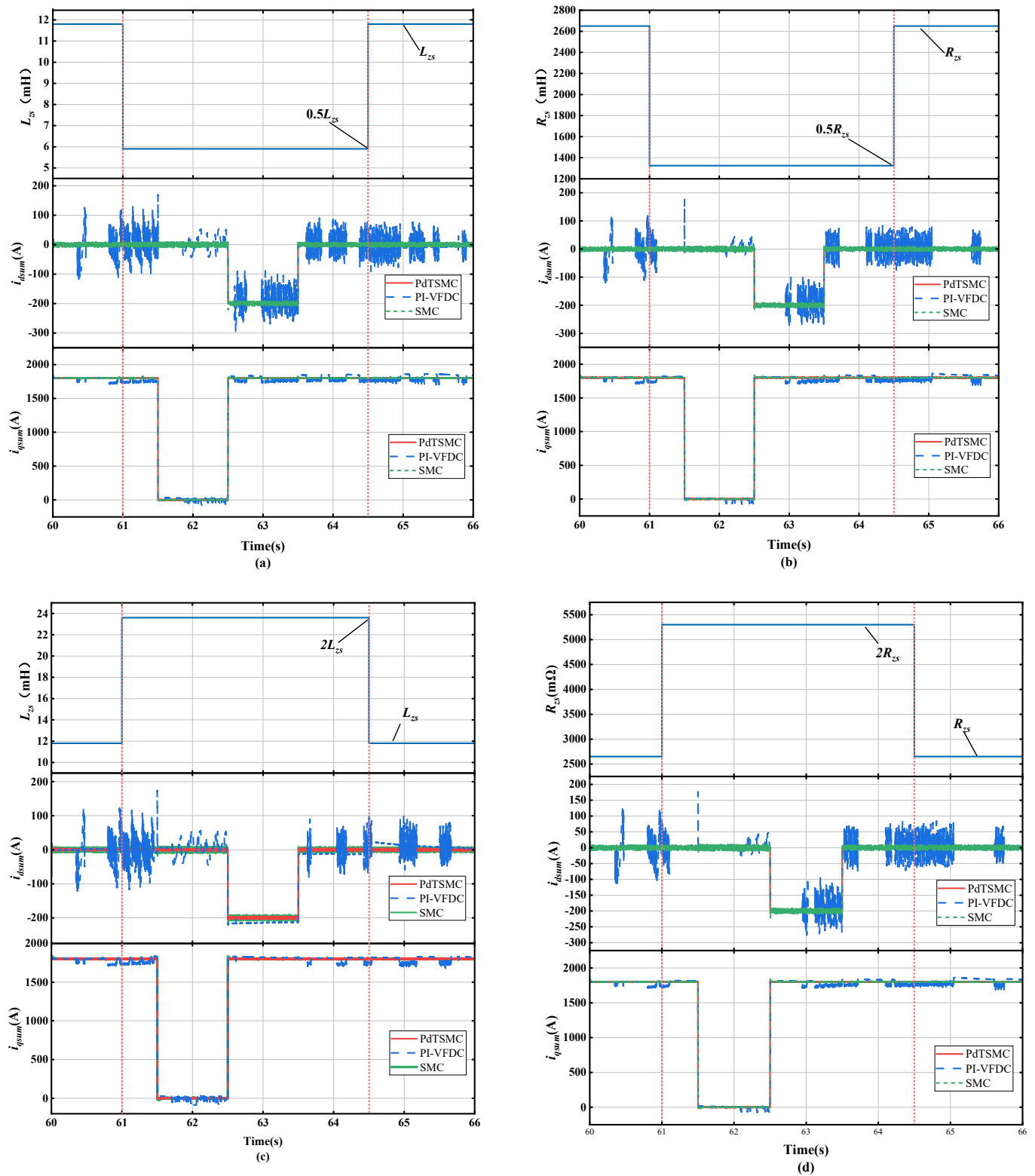


Figure 5. The dq-axis current curves under parameter perturbations: (a) $0.5 L_z$; (b) $0.5 R_z$; (c) $2 L_z$; (d) $2 R_z$.

Current mutations did not result in overshoot, with the maximum current error being 20 A. By introducing a power-integral term, the controller output became smoother and more continuous, effectively reducing high-frequency oscillations in the system and further suppressing the chattering phenomenon. The maximum response time was limited to within 0.1 s, meeting the predefined convergence time of the controller. There-

fore, the proposed strategy III is highly suitable for stator current control under complex operating conditions.

To further validate the effectiveness of the train's global control strategy, the train was commanded to track the target curve shown in Figure 3. In this study, to assess the impact of parameter perturbation on train operation control, double LSM parameter perturbation disturbance was applied at 40 s and 350 s to evaluate the robustness and effectiveness of the current control strategy. Considering that two high-power converters with the same power rating were connected in parallel to supply power to the motor, this paper further simulated and analyzed the voltage and current at the motor input. The simulation results showed that the Space Vector Pulse Width Modulation (SVPWM) output waveform, as shown in Figure 6, revealed variations in the three-phase voltage due to the magnification of the motor parameters, resulting in fluctuations in the modulation ratio. As the train was in the acceleration phase, the frequency of the three-phase current input increased, but its amplitude remained unchanged, ensuring the stable output of the maglev train's traction force. This phenomenon validates the inner-loop control strategy's ability to suppress system disturbances during dynamic operations, thereby proving the effectiveness and stability of the current control strategy.

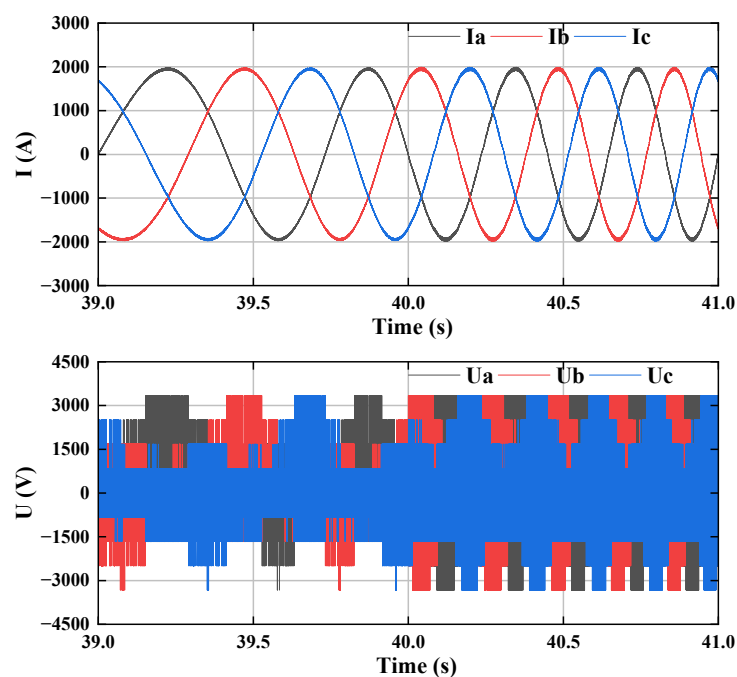


Figure 6. SVPWM output waveform.

To verify the robustness of the designed dual-loop control system, various types of disturbances were introduced in the simulation, and all relevant graphs were updated. A sinusoidal wave disturbance was added from 40 to 50 s, Gaussian white noise was introduced from 200 to 210 s, and a spike pulse disturbance was applied from 350 to 360 s. These disturbances allowed for a more comprehensive evaluation of the FTDO's compensation performance for lumped aggregate disturbances.

The global stator current response curve under the PdTSMC dual-loop control strategy with the finite-time disturbance observer is shown in Figure 7. During both the acceleration and cruising phases, the effects of parameter perturbations were nearly eliminated, further validating the robustness of the PdTSMC.

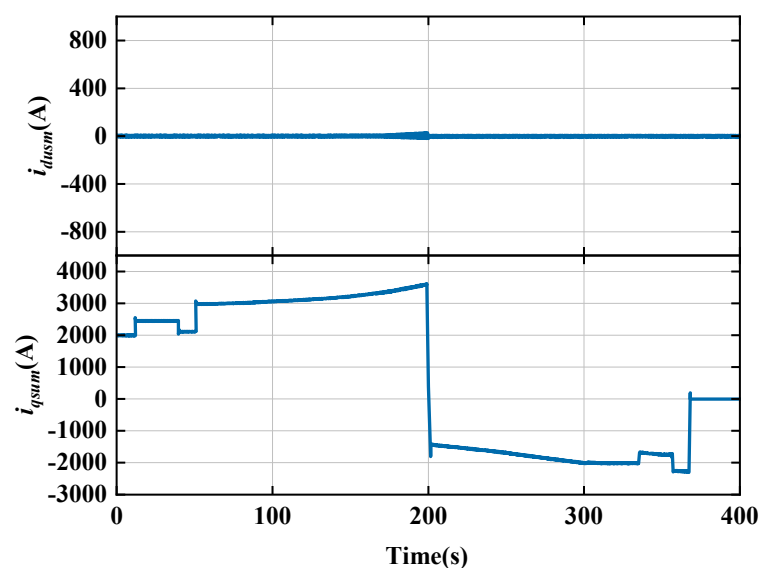


Figure 7. The full stator current response curve.

Specifically, the dual-end power supply system refers to four high-power modules simultaneously providing power to the vehicle. In this study, two high-power modules were connected in parallel to supply power to the long-stator linear synchronous motor on one side of the track. To accurately simulate this power supply method, the simulation assumed that the single-side motor handled double the load, thereby simulating the scenario where both sides of the motor simultaneously provided traction to the vehicle. This simplified assumption helped to theoretically reflect the power distribution characteristics of the dual-end power supply system and provides a more reasonable simulation model for subsequent control strategy validation [25].

In order to verify the tracking performance of the speed loop controller, a simulation analysis was conducted based on the target operation trajectory shown in Figure 3. The power of the two inverters was set to 24 MVA to limit the output amplitude of the current loop. The parameters of the speed loop controller are shown in Table 4.

To fully validate the effectiveness of the outer-loop control algorithm, six control algorithms were selected for comparative verification. The tracking performances of different outer-loop control algorithms are shown in Figures 8 and 9, and the maximum outer-loop controller performance is shown in Table 5.

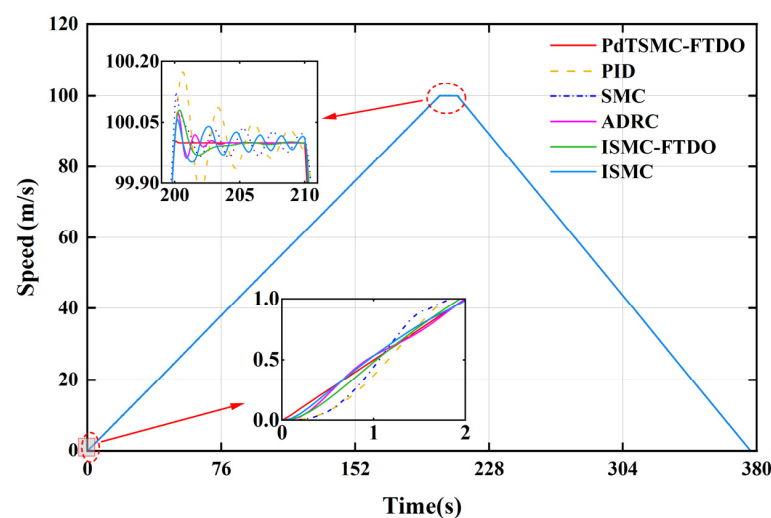


Figure 8. The full-speed tracking curve.

Table 4. The speed loop controller parameters.

Controller	Parameters	Value
PID	K_p	7
	K_i	20
	K_d	2
SMC	δ_1	18
	δ_2	8000
	K	55
ISM	c_1	5
	c_2	20
	k_1	5000
	k_2	5500
ISM-FTDO	c_1	5
	c_2	20
	k_1	5000
	k_2	5500
	β_0	100
	β_1	100
	β_2	100
	L	1
ADRC	α_1	0.5
	α_2	0.25
	Δ	0.1
	β_1	180
	β_2	2160
	β_3	24,000
PdTSMC-FTDO	λ	0.2
	T_1	1
	T_2	1
	c_3	10
	c_6	100
	β_0	100
	β_1	100
	β_2	100
	L	1

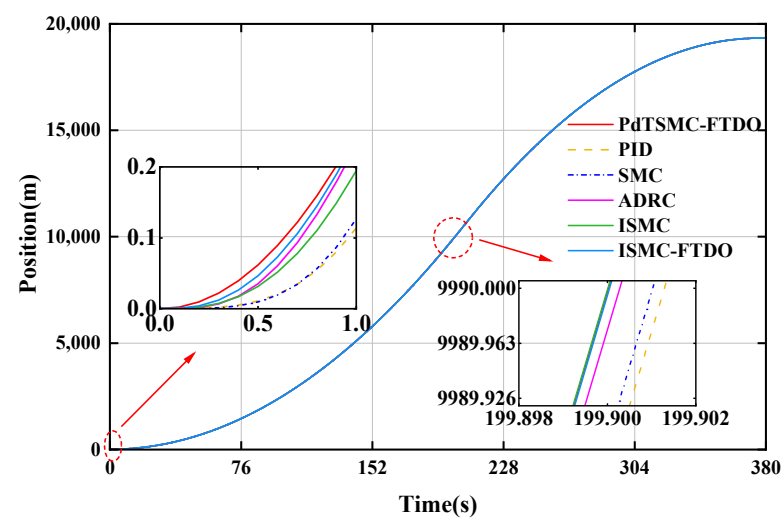
**Figure 9.** The full-position tracking curve.

Table 5. Comparison of outer-loop controller performance.

Controller	Maximum Speed Error	Maximum Position Error	Response Time
PID	0.25 m/s	0.15 m	30 s
SMC	0.18 m/s	0.13 m	10 s
ISM	0.098 m/s	0.023 m	6 s
ISM-FTDO	0.047 m/s	0.068 m	5 s
ADRC	0.084 m/s	0.029 m	3 s
PdTSMC	0.0005 m/s	0.0005 m	0.5 s

In the case of the same inner-loop controller, the PID controller exhibits the most significant oscillation, with the maximum tracking errors for speed and displacement being 0.25 m/s and 0.15 m, respectively. When faced with various nonlinear disturbances in both the inner and outer loops, the system's maximum response time is 30 s, displaying significant delay effects. This performance is insufficient to meet the high demands of real-time and robustness required for the control of high-speed maglev trains.

In contrast, the linear sliding mode controller (SMC) designed in reference [26] improves system performance to some extent. Specifically, the maximum tracking errors for speed and displacement are reduced to 0.18 m/s and 0.13 m, respectively, mitigating the impact of nonlinear disturbances. Moreover, the system's maximum response time is limited to under 10 s. However, the system still exhibits delay phenomena, and transient response optimization is needed, with steady-state errors still requiring improvement.

To comprehensively evaluate the control performance of the designed sliding mode controller, this study employed the integral sliding mode controller proposed in reference [27] as a comparison algorithm. The results indicate that after adopting this controller, the maximum tracking errors for speed and displacement are reduced to 0.098 m/s and 0.023 m, respectively. The maximum response time is 6 s, but chattering phenomena remain evident, and transient performance needs further improvement.

Building upon the integral sliding mode control strategy, this study introduced a finite-time disturbance observer (FTDO) for real-time feedforward disturbance compensation. The experimental results show that the maximum tracking error for speed is further reduced to 0.047 m/s while the tracking error for displacement is 0.068 m. The chattering phenomenon is improved, and the maximum response time is further shortened to 5 s.

Additionally, a comparison was made with the improved ADRC control strategy proposed in reference [28]. After adopting this strategy, the maximum speed tracking error is limited to within 0.084 m/s, the maximum displacement tracking error is reduced to 0.029 m, and the maximum response time is successfully shortened to under 3 s, further enhancing the transient performance.

Figures 10 and 11 display the control performance of the PdTSMC based on the finite-time disturbance observer. It can be observed that the tracking accuracy has significantly improved, with the maximum velocity and displacement tracking errors limited to within 0.0005 m/s and 0.0005 m, respectively. During the cruise phase, bias-free tracking is achieved. Under complex nonlinear disturbances, the velocity response time is 0.5 s, and the displacement response time is 0.2 s, which aligns with the pre-set convergence time of the outer-loop controller. The system demonstrates excellent steady-state and transient performance.

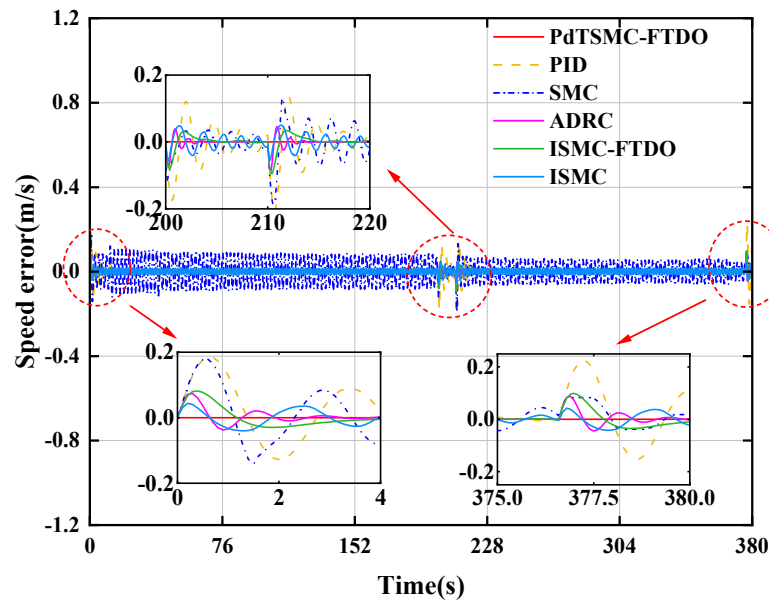


Figure 10. The speed tracking error.

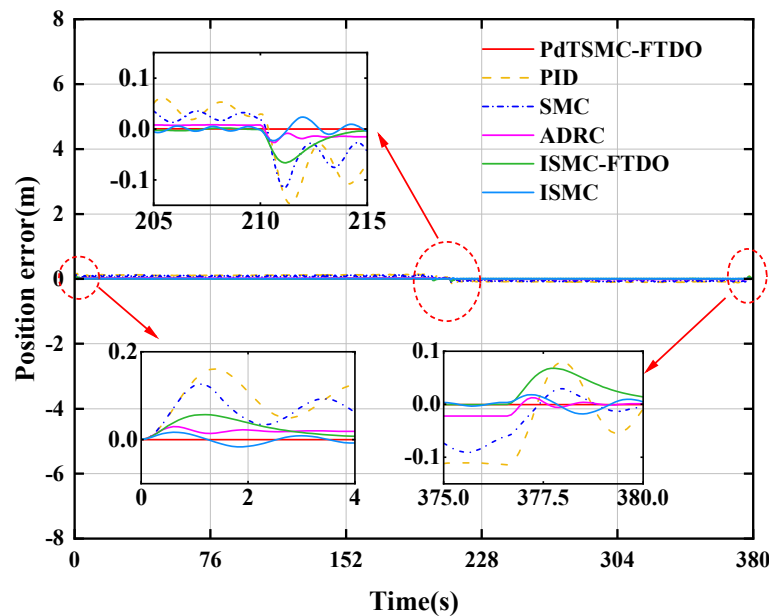


Figure 11. The position tracking error.

As shown in Figure 12, during the operation of the maglev train, the FTDO can effectively track and observe the dynamic changes of lumped disturbances in the system. When faced with various types of disturbance signals, the FTDO enhances the system's ability to rapidly estimate unknown disturbances through feedforward compensation of the system output. This compensation mechanism enables the system to respond quickly to disturbance changes during real-time operation, thereby improving the robustness and stability of the train's control system.

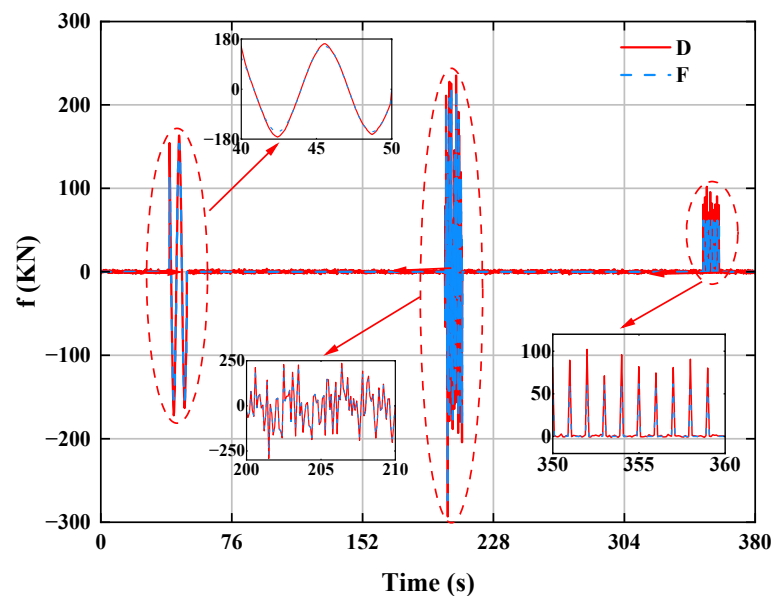


Figure 12. Lumped aggregate disturbance.

5. Conclusions

This paper addresses the control problem of a third-order high-speed maglev train under the influence of modeling inaccuracies and parameter perturbations and proposes a global finite-time Field-Oriented Control (FOC) method. Firstly, an improved sliding mode control structure was designed based on predefined-time stability theory, and a predefined-time sliding mode controller was applied to stator current, speed and displacement tracking control, enhancing the system's real-time performance and disturbance resistance. Furthermore, to address issues of inaccurate train modeling and external disturbance estimation, unknown disturbances and modeling uncertainties were normalized as lumped terms. A finite-time disturbance observer was designed for real-time feedforward compensation, ensuring global finite-time stability of the system while further enhancing robustness. Finally, through simulation analysis of the train's operation, the effectiveness of the proposed control strategy in terms of robustness and dynamic tracking performance was validated. The results show that the method improves the tracking performance of the maglev train in terms of position, speed and stator current and strengthens the robustness of the operation control system.

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